

LED Brightness Control Using Web Interface

Author: M Naga Hema

Abstract

This project demonstrates controlling LED brightness using Arduino UNO and a web-based interface. Users can control LED brightness in real time using a slider through Web Serial API communication.

Objectives

- Control LED brightness through browser interface.
- Demonstrate PWM and serial communication.
- Integrate hardware and software technologies.

Technologies Used

Arduino UNO, HTML, CSS, JavaScript, Web Serial API, PWM, Embedded Systems.

Hardware Components

- Arduino UNO
- LED
- Resistor
- Jumper Wires
- USB Cable

Working Principle

The user adjusts a slider on the web page. JavaScript sends brightness values to Arduino using Web Serial API. Arduino receives the values and changes LED brightness using PWM through `analogWrite()`.

Applications

- Smart Home Automation
- Mood Lighting
- Educational Electronics Projects
- IoT Lighting Systems

Advantages

- User-friendly interface
- Real-time control
- Easy scalability
- Interactive learning platform

Future Enhancements

- Mobile application integration
- Voice assistant support
- IoT cloud integration
- RGB LED support

Conclusion

The project successfully demonstrates browser-controlled LED brightness using Arduino UNO and Web Serial API. It combines web development with embedded systems and provides a strong foundation for future IoT-based applications.